

ICAD RT-4 · Set 3 — Answer Sheet

Exam 12-10-2026 · 75 Q · 300 marks · 3 hours · +4 correct, -1 wrong, 0 blank

Name

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Physics

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Mathematics

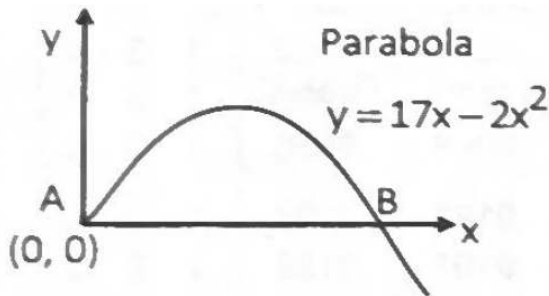
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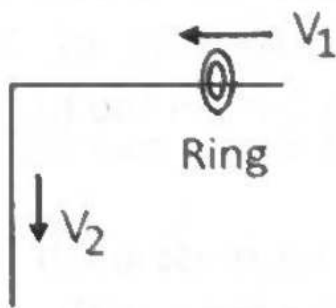
Physics

PART A · Q1–20 · SINGLE CORRECT

1. P1L3V3Q4 Distance between A and B (x-intercepts) for parabola $y = 17x - 2x^2$. (a) 15.5 units (b) 16.5 units (c) 8.5 units (d) 12.5 units



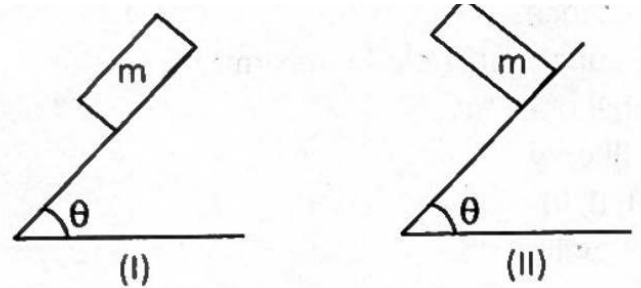
2. P2L2V4Q2 $U = k(r-r_0)^2$, find $F = -dU/dr$ ($-2k(r-r_0)$). (a) $-k(r-r_0)$ (b) $-2k(r-r_0)$ (c) $-k(r_0-r)$ (d) $k(r-r_0)$
3. P3L2V2Q2 Find the resultant of two perpendicular vectors $A = 9$ N and $B = 12$ N. (a) 15 N (b) 13 N (c) 10 N (d) 20 N
4. P4L2V4Q5 A body moves for 9 s total, starting from rest with uniform acceleration then uniform retardation (twice the acceleration). Time during retardation: (a) 3 s (b) 4.5 s (c) 5 s (d) 6 s
5. P4L4V3Q5 A body dropped from a height h , with initial speed zero, strikes the ground with velocity 3 km/h. Another body of the same mass is dropped from the same height h with an initial speed $-u' = 4$ km/h. Find the final velocity of the second body with which it strikes the ground. (a) 3 (b) 4 (c) 5 (d) 12 km/h
6. P4L8V4Q5 Body projected at 30° with speed 30 ms^{-1} . Angle with the horizontal after 1.5 seconds? (a) 0 deg (b) 30 deg (c) 60 deg (d) 90 deg
7. P4L10V6Q8 A plane is inclined at angle θ . A bullet is fired from the bottom with velocity v . Find the maximum possible range on the inclined plane. ($v^2/g(1+\sin\theta)$). (a) v^2/g (b) $v^2/g(1+\sin\theta)$ (c) $v^2/g(1-\sin\theta)$ (d) $v^2/g(1+\cos\theta)$
8. P5L2V3Q4 Two masses $m_1 = 3$ kg and $m_2 = 5$ kg are connected by a massless string over a frictionless pulley; the acceleration of the system is: (a) 1.25 m/s^2 (b) 2.5 m/s^2 (c) 5 m/s^2 (d) 10 m/s^2
9. P6L1V3Q3 A ring of mass 100 g changes velocity from $V_1 = -7\hat{i}$ to $V_2 = -4\hat{j}$ in 1 s. Average force by the frame? (a) 0.806 N (b) 8.06 N (c) 80.6 N (d) 806 N



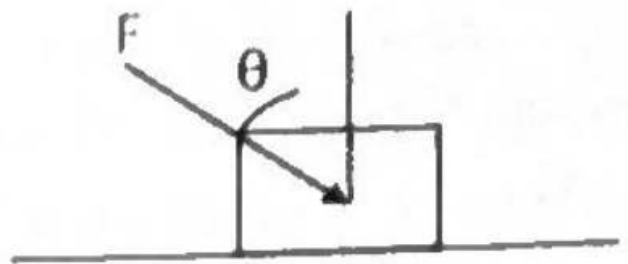
10. P6L2V5Q4 Arrangement 4 kg, 2 kg, 1 kg by light inextensible strings, smooth, $F=28$ N. Acceleration of each block. (a) 8 m/s^2 (b) 4 m/s^2 (c) 2 m/s^2 (d) 14 m/s^2



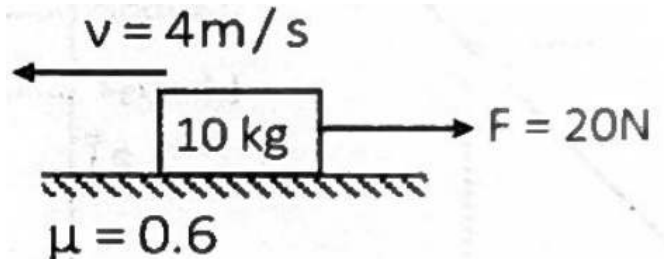
11. P7L1V3Q4 A block is first placed on its long side and then on its short side on the same inclined plane. The block slides down in situation (II) but remains at rest in situation (I). A possible explanation is: (a) Normal force is less in situation (II) (b) The frictional force is less in situation II because the contact area is less (c) The shorter side is smoother (d) In situation I the frictional force is



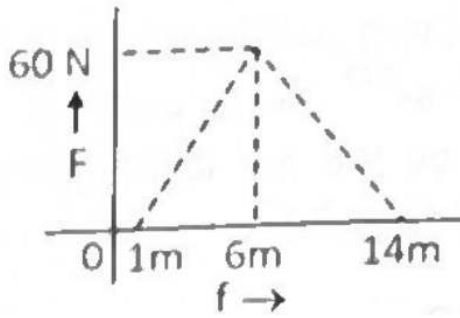
12. P7L2V5Q1 A block of mass M is placed on a rough horizontal surface. A force $F = mg$ acts on the block, inclined to the vertical at an angle θ (pushing down and forwards). The coefficient of friction is μ . The block can be pushed along the surface only if: (a) $\tan\theta \geq \mu$ (b) $\cot\theta \geq \mu$ (c) $\tan(\theta/2) \geq \mu$ (d) $\cot(\theta/2) \geq \mu$



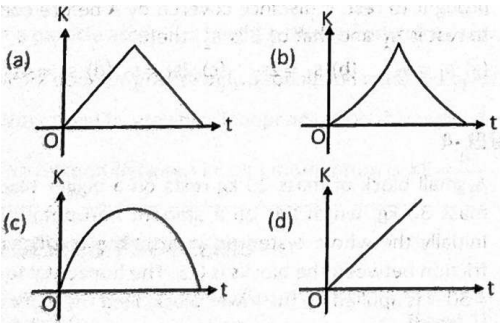
13. P7L2V5Q4 A block of mass 10 kg is moving on a rough surface with $v = 4 \text{ m/s}$ while a force $F = 20$ N acts on it as shown in the figure; $\mu = 0.6$. The frictional force acting on the block is ($g = 10 \text{ m/s}^2$): (a) 20 N (b) 40 N (c) 60 N (d) 80 N



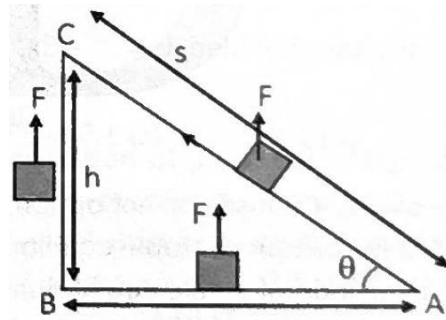
14. P8L2V5Q1 A force acts on a body and displaces it in its direction. The graph shows the relation between the force and displacement; the work done by the force is: [Graph: $F = 0$ at 1 m, rises linearly to a peak of 60 N at 6 m, then falls linearly back to 0 at 14 m.] (a) 420 J (b) 390 J (c) 840 J (d) 720 J



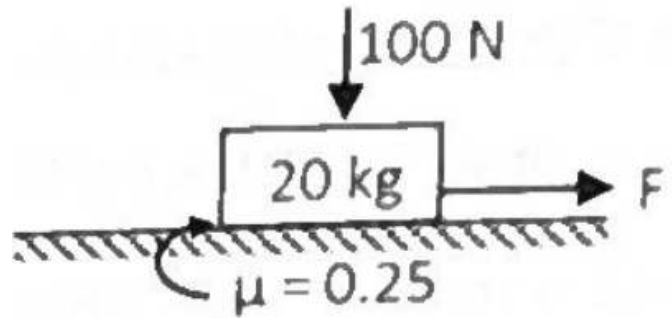
15. P8L3V4Q6 A ball is dropped at $t = 0$ from a height on a smooth elastic surface. Identify the graph which correctly represents the variation of kinetic energy K with time t . (options: (a) triangular straight-line rise and fall, (b) concave-up rise to a sharp peak then concave-up fall, (c) dome/inverted-parabola, (d) straight line increasing)



16. P8L5V3Q2 Path AC or ABC: a block is taken from A to C, either straight up the incline AC (length s , angle θ at A) or via the horizontal leg AB and then the vertical leg BC (height h). A constant vertical force F (drawn upward in the figure) acts on the block throughout. Then (a) the work done by the force F for both paths will be same (b) $W_{AC} > W_{ABC}$ (c) $W_{AC} < W_{ABC}$ (d) none of the above



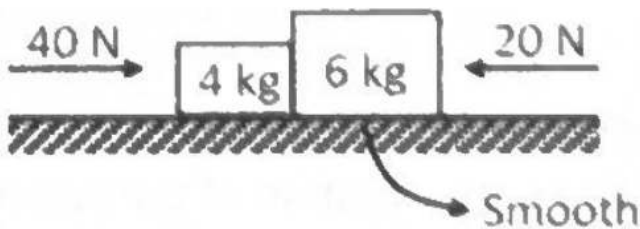
17. P4L8V3Q3 Ball projected on a planet: $x = 10\sqrt{3}t$, $y = 10t - t^2$. Maximum height attained? (a) 100 m (b) 75 m (c) 50 m (d) 25 m
18. P7L2V4Q1 Find F_{max} (in newtons) so that the block remains at rest. A 20 kg block on a rough floor ($\mu = 0.25$) is pressed by a 100 N force acting vertically downwards on its top, while a horizontal force F is applied. ($g = 10 \text{ m/s}^2$) (write the value)



19. P8L8V3Q4 A motor pump set of efficiency 80 %, lifts 800 litres of water in 19.6 seconds over a head of 20 m. Its input power is (a) 64 kW (b) 40 kW (c) 10 kW (d) 196 kW
20. P8L1V3Q1 A man moves on a straight horizontal road with a block of mass 2 kg in his hand. If he covers a distance of 40 m with an acceleration of 0.5 m/s^2 , find the work done (in J) by the man on the block during the motion. (write the value)

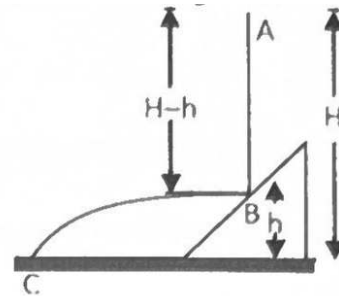
PART B · Q21–25 · NUMERICAL VALUE

21. P6L1V6Q10 Find the normal reaction between the 4 kg and 6 kg blocks with 40 N and 20 N applied from opposite sides on a smooth surface.

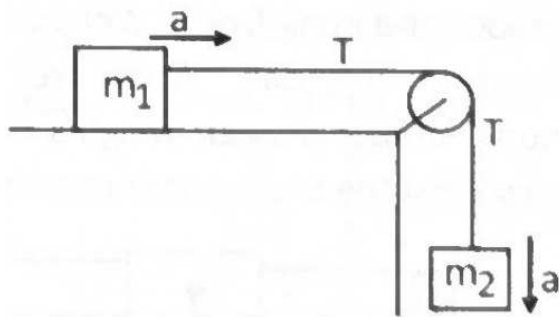


22. P3L1V4Q2 A 300 gm mass has a velocity of $\vec{v} = 3\hat{i} + 4\hat{j} \text{ m/s}$ at a certain instant. Its kinetic energy is ____.

23. P4L3V6Q6 A body falling freely from a given height H hits an inclined plane in its path at a height h . As a result of this impact, the direction of the velocity becomes horizontal. For what value of h/H will the body take maximum time to reach the ground?



24. P6L3V5SuggQ2 A block m_1 resting on a smooth table is connected to a mass m_2 through a light pulley. Find the tension in the string.



25. P3L5V5Q4 A particle moves from position $3\hat{i}+2\hat{j}-6\hat{k}$ to $14\hat{i}+13\hat{j}+9\hat{k}$ due to a uniform force $(4\hat{i}+\hat{j}+3\hat{k})\text{N}$; displacement in m. Find the work done, in joule.

Chemistry

PART A · Q1–20 · SINGLE CORRECT

- C2L3V4Q3 Photon of wavelength $12.5 \text{ \AA}$ — calculate its energy in J and eV. (a) $9.91 \times 10^{-17} \text{ J} = 6.18 \times 10^{-2} \text{ eV}$ (b) $1.23 \times 10^{-16} \text{ J} = 7.71 \times 10^{-2} \text{ eV}$ (c) $2.35 \times 10^{-16} \text{ J} = 1.46 \times 10^{-3} \text{ eV}$ (d) $1.58 \times 10^{-16} \text{ J} = 9.92 \times 10^{-2} \text{ eV}$
- C2L4V3Q1 Radius of the 2nd orbit of Li^{2+} ($Z=3$), $a_0 = 5.29 \times 10^{-11} \text{ m}$. (a) $7.05 \times 10^{-11} \text{ m}$ (b) $1.18 \times 10^{-10} \text{ m}$ (c) $2.12 \times 10^{-10} \text{ m}$ (d) $5.29 \times 10^{-11} \text{ m}$
- C2L4V4Q1 Condition an electron must satisfy for an allowed orbit. (a) $mvr = nh/\pi$ (b) mvr varies continuously with radius (c) $mvr = nh/2\pi$ (d) $mvr = nh/4\pi$
- C2L8V6Q5 0.25 kg particle, $\Delta x = 10^{-5}$ — uncertainty in velocity. (a) $1.2 \times 10^{-34} \text{ m/sec}$ (b) $2.1 \times 10^{-29} \text{ m/sec}$ (c) $1.6 \times 10^{-20} \text{ m/sec}$ (d) $1.7 \times 10^{-9} \text{ m/sec}$
- C3L1V3Q3 Formula mass of MgCl_2 ($\text{Mg}=24$, $\text{Cl}=35.5$). (a) 94 u (b) 95 u (c) 60 u (d) 70 u
- C3L2V3Q5 Calculate the number of electrons in 0.5 mol of water (each $\text{H}_2\text{O} = 10$ electrons). (a) 5.011×10^{24} (b) 3.011×10^{23} (c) 6.022×10^{24} (d) 3.011×10^{24}
- C3L3V3Q4 A gas has vapour density 20. What is its molecular mass and density at STP? (a) 40 g/mol , 1.79 g/L (b) 20 g/mol , 0.89 g/L (c) 40 g/mol , 2.5 g/L (d) 10 g/mol , 0.50 g/L
- C3L3V5Q9 11.2 L of a gas at STP weighs 14 g . The gas could be: (a) N_2 (b) CO (c) NO_2 (d) N_2O
- C3L5V6Q3 10 g mixture of NaHCO_3 and Na_2CO_3 has 1.68 g NaHCO_3 . It is heated at 400 K . The weight of the residue will be. (a) 9.38 g (b) 8.32 g (c) 10.0 g (d) 1.68 g
- C4L3V3Q1 Calculate radius difference between O (66 pm) and O^{2-} (140 pm). (a) 74 pm (b) 84 pm (c) 70 pm (d) 80 pm
- C5L1V4Q4 In which compound is the ionic character expected to be maximum? (a) SrCl_2 (b) BaCl_2 (c) CaCl_2 (d) CsCl
- C5L4V5SuggQ4 Hybridisation state of Xe in XeF_2 , XeF_4 and XeF_6 respectively are: (a) sp^2 , sp^3d , sp^3d^2 (b) sp^3d , sp^3d^2 , sp^3d^3 (c) sp^3d^2 , sp^3d , sp^3d^3 (d) sp^2 , sp^3 , sp^3d
- C5L7V5Q2 Molecular orbital theory is preferred over valence bond theory as it can explain: (a) bond strength (b) geometry of molecule (c) the magnetic nature of the molecule (d) none of these
- C5L10V3Q2 Which of the following correctly explains why ethanol is soluble in water while ethane is not? (a) Ethanol has higher molecular mass (b) Ethanol can form hydrogen bonds with water (c) Ethane has non-polar C—H bonds (d) Ethanol has shorter bond length
- C5L10V3Q4 For a covalent molecule AB, the bond length is correctly given by: (a) Difference of covalent radii of A and B (b) Average of atomic radii of A and B (c) Sum of covalent radii of A and B (d) Sum of van der Waals radii of A and B
- C1L9V2Q1 Calculate T_c if $a=3.6 \text{ atm}\cdot\text{L}^2/\text{mol}^2$, $b=0.0427 \text{ L/mol}$ ($R=0.0821$). (a) 295 K (b) 304 K (c) 312 K (d) 328 K
- C2L10V4Q2 $\text{EC } 1s^2 2s^2 2p^3$ — number of unpaired electrons. (a) 1 (b) zero (c) 3 (d) 5
- C4L4V3Q3 If IE of element A is 13.6 eV , convert to kJ/mol . (a) 1312 (b) 1250 (c) 1400 (d) 1100
- C5L3V3Q3 Predict hybridisation of B in BF_3 and number of unhybridised p orbitals. (a) sp^2 , 1 (b) sp^3 , 0 (c) sp , 2 (d) dsp^2 , 0
- C1L6V5Q10 Average KE per molecule of an ideal gas at 27°C (1 atm). (a) $900 \text{ cal}\cdot\text{mol}^{-1}$ (b) $6.2 \times 10^{-21} \text{ J}\cdot\text{molecule}^{-1}$ (c) $336.7 \text{ J}\cdot\text{mol}^{-1}$ (d) $3.74 \times 10^{-3} \text{ J}\cdot\text{mol}^{-1}$

PART B · Q21–25 · NUMERICAL VALUE

- C1L11V5Q2 Among CH_3OH , $\text{C}_4\text{H}_9\text{OH}$, CH_3OCH_3 — lowest b.p. and why? (a) CH_3OH - smallest size (b) CH_3OCH_3 - lacks intermolecular H-bonding (c) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ - largest chain (d) All have similar forces
- C4L4V6Q6 An element has successive ionization enthalpies of 940 (first), 2080 , 3090 , 4140 , 7030 , 7870 , 16000 and $19500 \text{ kJ mol}^{-1}$. To which group of the periodic table does this element belong?
- C3L1V4Q4 Boron mass 10.80 ; 80% isotope = 11.00 ; find the other isotope
- C1L3V3Q2 Density of gas ($M=28$) at 2 atm & 300 K . (a) 1.87 g/L (b) 2.00 g/L (c) 1.53 g/L (d) 2.27 g/L
- C1L4V4Q7 CO & N_2 equal masses, total 6 atm — ratio of partial pressures. (a) $1 : 1$ (b) $1 : 2$ (c) $2 : 1$ (d) $3 : 1$

Mathematics

PART A · Q1–20 · SINGLE CORRECT

- M1L1V3Q5 Angles of quadrilateral in ratio $2 : 5 : 8 : 9$; in radian. (a) $\pi/6$, $5\pi/12$, $3\pi/2$, $3\pi/4$ (b) $\pi/3$, $5\pi/12$, $2\pi/3$, $2\pi/5$ (c) $\pi/6$, $5\pi/12$, $2\pi/3$, $4\pi/3$ (d) $\pi/6$, $5\pi/12$, $2\pi/3$, $3\pi/4$
- M2L3V2Q4 A = letters of "ACCOUNTANCY" → cardinality of A. (a) 5 (b) 6 (c) 7 (d) 8

- 3.** M3L3V2Q4 $A=130^\circ$, $x=\sin A+\cos A \rightarrow$ sign of x . (a) $x > 0$ (b) $x < 0$ (c) $x = 0$ (d) $x \geq 0$
- 4.** M3L3V4Q10 Assertion $\sqrt{(1-\sin^2 100^\circ)} \cdot \sec 100^\circ - 1 = -2$. (a) A, R are true and R is correct explanation of A (b) A, R are true and R is not correct explanation of A (c) A is true, R is false (d) A is false, R is true
- 5.** M5AL1V5Q1 Find the value of $\cot(1965^\circ)$. (a) $2 + \sqrt{3}$ (b) $2 - \sqrt{3}$ (c) $\sqrt{3} - 2$ (d) $-2 - \sqrt{3}$
- 6.** M5BL1V2Q5 $2 \cos((P+Q)/2) \cos((P-Q)/2) = ?$ (a) $\cos P + \cos Q$ (b) $\cos P - \cos Q$ (c) $\sin P + \sin Q$ (d) $\sin P - \sin Q$
- 7.** M5CL2V4Q4 For all real values of x , $\cot x - 2 \cot 2x$ is equal to (a) $\tan 2x$ (b) $\tan x$ (c) $-\cot 3x$ (d) none
- 8.** M6L2V4Q8 If $1/(x+2) < 1/4$, then $x \in ?$ (a) $(2, \infty)$ (b) $(-2, \infty)$ (c) $(-\infty, -2) \cup (2, \infty)$ (d) $(0, 2)$
- 9.** M6L5V4Q10 If $(x^2 - 5x + 6)/(x^2 + x + 1) < 0$, then x belongs to: (a) $(0, 3)$ (b) $(3, \infty)$ (c) $(2, 3)$ (d) $(-\infty, 2) \cup (3, \infty)$
- 10.** M7L1V4Q4 If $((x+1)^2(x-1)^3) / (x^4(x-2)) = 0$, then the sum of all roots is: (a) 0 (b) 1 (c) -1 (d) 2
- 11.** M7L1V5Q1 If $((6x-5)(x^2-4x)) / ((4x+1)x) = 0$, then the value of x : (a) $5/6, 4$ (b) $5/6, 4, 0$ (c) $5/6, 4, 0, -1/4$ (d) $5/6$
- 12.** M8AL3V3Q3 For an A.P. if $S_n = 2n^2 - 3n$ then value of t_{25} is: (a) 91 (b) 95 (c) 99 (d) 103
- 13.** M8BL2V1Q4 Which condition must be met for an infinite G.P. to have a finite sum? (a) $r > 1$ (b) $r = 1$ (c) $|r| < 1$ (d) $r = 0$
- 14.** M8BL2V3Q2 Find sum of first ten terms of the G.P. 3, -6, 12, -24, (write the value)
- 15.** M8CL1V3Q4 The harmonic mean of the roots of the equation $(5 + \sqrt{2})x^2 - (4 + \sqrt{2})x + 8 + 2\sqrt{2} = 0$ is (a) 2 (b) 4 (c) 6 (d) 8
- 16.** M5AL2V5Q6 $(\tan A + \tan B)/\tan(A+B) + (\tan A - \tan B)/\tan(A-B) = ?$ (a) 0 (b) 1 (c) 2 (d) $1/2$
- 17.** M6L1V3Q5 If $30x < 200$ & $x \in \mathbb{N}$, then the sum of all possible values of x is: (write the value)
- 18.** M4L3V3Q4 $f(x) = \text{sgn}(x)$ (write the value)
- 19.** M5BL2V5Q7 If $2 \sin(5\pi/12) \sin(\pi/12)$ is equal to k , then the value of k^2 is _____. (write the value)
- 20.** M8AL3V4Q3 A man starts repaying a loan as first instalment of Rs. 100. If he increases the instalments by Rs. 5 every month, what amount he will pay in the 30 th instalment? (write the value)

PART B · Q21–25 · NUMERICAL VALUE

- 21.** M2L2V6Q6 123 workers: ice-cream 42, tea 36, cold drink 30, 15/10/10 pairs, 11 IC&tea not CD $\rightarrow$ purchased nothing
- 22.** M3L1V5Q6 $(2\sin\theta\cos\theta - \cos\theta)/(1 - \sin\theta + \sin^2\theta - \cos^2\theta) = \lambda \rightarrow \lambda \cdot \tan\theta$
- 23.** M4L2V5Q4 Range of $f(x) = (2x-1)/(1-x) = R - \{a\} \rightarrow -2a+1$
- 24.** M2L1V3Q3 $A = \{y : y = 3x+1, 2 \leq x \leq 5\}$; subsets 2^n ; find $4n^2 - 15$
- 25.** M4L2V4Q3 Minimum value of $f(x) = x^2 - 4x + 5$